

Ejercicios de tcpdump

Es una herramienta de **captura y análisis de tráfico de red** en modo texto (CLI). Sirve para **ver paquetes que pasan por una interfaz de red** en tiempo real (Ethernet/Wi-Fi), y es muy usada en administración de redes y ciberseguridad.

♦ Ejercicios básicos

1. Listar interfaces disponibles

tcpdump -D

```
(kali㉿kali)-[~]  
$ tcpdump -D  
1.eth0 [Up, Running, Connected]  
2.eth1 [Up, Running, Connected]  
3.any (Pseudo-device that captures on all interfaces) [Up, Running]  
4.lo [Up, Running, Loopback]  
5.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]  
6.nflog (Linux netfilter log (NFLOG) interface) [none]  
7.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]  
8.dbus-system (D-Bus system bus) [none]  
9.dbus-session (D-Bus session bus) [none]
```

👉 Muestra todas las interfaces de red que tcpdump puede usar.

2. Capturar tráfico en la interfaz eth0

tcpdump -i eth0

```
(kali㉿kali)-[~]  
$ sudo tcpdump -i eth0  
  
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode  
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes  
05:29:45.531755 ARP, Request who-has 192.168.0.231 (Broadcast) tell 192.168.0.9  
0, length 46  
05:29:45.558708 ARP, Request who-has 192.168.0.232 (Broadcast) tell 192.168.0.9
```

👉 Ver los primeros 20 paquetes que pasan por eth0.

3. Guardar captura en un fichero

tcpdump -i eth0 -w trafico.log

👉 Luego ábrelo con:

tcpdump -r trafico.log

```
(kali@kali)-[~]
$ sudo tcpdump -i eth0 -w trafico.log

tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C1985 packets captured
1985 packets received by filter
0 packets dropped by kernel

(kali@kali)-[~]
$ tcpdump -r trafico.log

reading from file trafico.log, link-type EN10MB (Ethernet), snapshot length 262144
05:30:36.142676 ARP, Request who-has 192.168.1.174 tell 192.168.0.77, length 46
05:30:36.182943 IP 192.168.0.97.mdns > mdns.mcast.net.mdns: 0 ANY (QM)? DESKTOP-M7CM9BE._dosvc._tcp.local. (51)
05:30:36.185001 IP6 fe80::d12d:c7e2:4ae3:1684.mdns > ff02::fb.mdns: 0 ANY (QM)? DESKTOP-M7CM9BE._dosvc._tcp.local. (51)
05:30:36.198699 ARP, Request who-has 192.168.1.176 tell 192.168.0.48, length 46
05:30:36.242183 IP 192.168.248.2 > all-systems.mcast.net: igmp query v2
05:30:36.242186 IP6 fe80::a697:33ff:fe4e:9fc4 > ip6-allnodes: HBH ICMP6, multicast listener querymax resp delay: 1000 addr: ::, length 24
05:30:36.244459 IP 192.168.0.108 > mdns.mcast.net: igmp v2 report mdns.mcast.net
05:30:36.244461 IP 192.168.0.108 > 239.255.255.250: igmp v2 report 239.255.255.250
```

◆ Filtrado por protocolos y puertos

4. Capturar solo tráfico TCP en eth0

tcpdump -i eth0 tcp

```
(kali@kali)-[~]
$ sudo tcpdump -i eth0 tcp

tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
05:32:30.562886 IP kali.33728 > 203.137.36.34.bc.googleusercontent.com.https: Flags [S], seq 538498950, win 64240, options [mss 1460,sackOK,TS val 2276037332 ecr 0,nop,wscale 7], length 0
05:32:30.579807 IP 203.137.36.34.bc.googleusercontent.com.https > kali.33728: Flags [S], seq 2186694783, ack 538498951, win 65535, options [mss 1412,sackOK,TS val 991401380 ecr 2276037332,nop,wscale 8], length 0
05:32:30.579913 IP kali.33728 > 203.137.36.34.bc.googleusercontent.com.https: Flags [A], ack 1, win 502, options [nop,nop,TS val 2276037348 ecr 991401380], length 0
05:32:30.582496 IP kali.33728 > 203.137.36.34.bc.googleusercontent.com.https: Flags [P], seq 1:677, ack 1, win 502, options [nop,nop,TS val 2276037352 ecr 991401380], length 676
05:32:30.590800 IP kali.33728 > 203.137.36.34.bc.googleusercontent.com.https: Flags [F], seq 677, ack 1, win 502, options [nop,nop,TS val 2276037365 ecr 991401380], length 0
```

5. Capturar tráfico en el puerto 80 (HTTP)

tcpdump -i eth0 tcp port 80

```
(kali@kali)-[~]
$ sudo tcpdump -i eth0 tcp port 80

tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
05:33:00.805492 IP kali.34244 > 82.221.107.34.bc.googleusercontent.com.http: Flags [S], seq 8222110734, win 64240, options [mss 1460,sackOK,TS val 2276037332 ecr 0,nop,wscale 7], length 0
05:33:00.821530 IP 82.221.107.34.bc.googleusercontent.com.http > kali.34244: Flags [S], seq 2186694783, ack 8222110734, win 65535, options [mss 1412,sackOK,TS val 991401380 ecr 2276037332,nop,wscale 8], length 0
05:33:00.821558 IP kali.34244 > 82.221.107.34.bc.googleusercontent.com.http: Flags [A], ack 1, win 502, options [nop,nop,TS val 2276037348 ecr 991401380], length 0
05:33:00.821900 IP 82.221.107.34.bc.googleusercontent.com.http > kali.34244: Flags [P], seq 1:677, ack 1, win 502, options [nop,nop,TS val 2276037352 ecr 991401380], length 676
05:33:00.837128 IP 82.221.107.34.bc.googleusercontent.com.http > kali.34244: Flags [F], seq 677, ack 1, win 502, options [nop,nop,TS val 2276037365 ecr 991401380], length 0
05:33:00.837914 IP 82.221.107.34.bc.googleusercontent.com.http > kali.34244: Flags [P], seq 1:677, ack 1, win 502, options [nop,nop,TS val 2276037365 ecr 991401380], length 676
05:33:00.837933 IP kali.34244 > 82.221.107.34.bc.googleusercontent.com.http: Flags [A], ack 1, win 502, options [nop,nop,TS val 2276037365 ecr 991401380], length 0
05:33:02.442783 IP kali.47810 > muc11s11-in-f3.1e100.net.http: Flags [S], seq 418723753, win 64240, options [mss 1460,sackOK,TS val 2276037365 ecr 0,nop,wscale 7], length 0
05:33:02.443087 IP muc11s11-in-f3.1e100.net.http > kali.47810: Flags [S], seq 3357292474, win 64240, options [mss 1460,sackOK,TS val 2276037365 ecr 0,nop,wscale 7], length 0
```

6. Capturar tráfico DNS (UDP puerto 53)

tcpdump -i eth0 udp port 53

```
(kali@kali)-[~]
$ sudo tcpdump -i eth0 udp port 53

tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
05:33:56.986237 IP kali.55606 > 254.red-80-58-61.staticip.rima-tde.net.domain: 25201+ A? contile.services.mozilla.com. (46)
05:33:56.986342 IP kali.55606 > 254.red-80-58-61.staticip.rima-tde.net.domain: 48501+ AAAA? contile.services.mozilla.com. (46)
05:33:57.008214 IP 254.red-80-58-61.staticip.rima-tde.net.domain > kali.55606: 25201 1/0/0 A 34.36.137.203 (62)
05:33:57.008214 IP 254.red-80-58-61.staticip.rima-tde.net.domain > kali.55606: 48501 0/1/0 (127)
```

- ◆ Filtrado por IP

7. Tráfico desde una IP concreta (host origen)

tcpdump -i eth0 src 192.168.43.143

```
(kali㉿kali)-[~]  
$ sudo tcpdump -i eth0 src 192.168.0.110  
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode  
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes  
05:36:36.432261 IP kali.44165 > 254.red-80-58-61.staticip.rima-tde.net.domain: 40591+ A? contile.services.mozilla.com. (46)  
05:36:36.432366 IP kali.44165 > 254.red-80-58-61.staticip.rima-tde.net.domain: 3981+ AAAA? contile.services.mozilla.com. (46)  
05:36:36.461110 IP kali.34412 > 203.137.36.34.bc.googleusercontent.com.https: Flags [S], seq 3278402167, win 64240, options [ms  
05:36:36.476495 IP kali.34412 > 203.137.36.34.bc.googleusercontent.com.https: Flags [A], seq 4233521686, win 502, options [nop
```

8. Tráfico hacia una IP concreta (host destino)

tcpdump -i eth0 dst 192.168.43.143

```
(kali㉿kali)-[~]  
$ sudo tcpdump -i eth0 dst 192.168.0.130  
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode  
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes  
05:37:45.316022 IP kali > 192.168.0.130: ICMP echo request, id 3, seq 1, length 64  
05:37:46.317686 IP kali > 192.168.0.130: ICMP echo request, id 3, seq 2, length 64  
05:37:47.319912 IP kali > 192.168.0.130: ICMP echo request, id 3, seq 3, length 64  
05:38:44.386051 ARP, Request who-has 192.168.0.130 (Broadcast) tell 192.168.0.90, length 46
```

9. Tráfico de y hacia una IP concreta (host completo)

tcpdump host 192.168.1.143

```
(kali㉿kali)-[~]  
$ sudo tcpdump host 192.168.0.130
```

- ◆ Filtrado por MAC y red

10. Tráfico con destino a una dirección MAC

tcpduimp ether dst 8A:B1:11:A0:BC:53

```
(kali㉿kali)-[~]
└─$ sudo tcpdump ether dst 08:00:27:d1:f8:5d
[sudo] password for kali:
Sorry, try again.
[sudo] password for kali:
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
06:04:01.744902 IP 254.red-80-58-61.staticip.rima-tde.net.domain > kali.38666: 1253
.137.203 (62)
06:04:01.744903 IP 254.red-80-58-61.staticip.rima-tde.net.domain > kali.38666: 41703
06:04:01.791019 IP 254.red-80-58-61.staticip.rima-tde.net.domain > kali.38659: 34202
4.red-80-58-61.staticip.rima-tde.net. (95)
06:04:01.794061 IP 203.137.36.34.bc.googleusercontent.com.https > kali.59227: UDP, l
06:04:01.811866 IP 203.137.36.34.bc.googleusercontent.com.https > kali.59227: UDP, l
06:04:01.820409 IP 254.red-80-58-61.staticip.rima-tde.net.domain > kali.41858: 53564
mc.prod.ads.prod.webservices.mozgcp.net., A 34.36.137.203 (106)
06:04:01.820410 IP 254.red-80-58-61.staticip.rima-tde.net.domain > kali.41858: 26687
mc.prod.ads.prod.webservices.mozgcp.net. (180)
06:04:01.831132 IP 203.137.36.34.bc.googleusercontent.com.https > kali.59227: UDP, l
06:04:01.859842 IP 203.137.36.34.bc.googleusercontent.com.https > kali.39342: UDP, l
06:04:01.859844 IP 203.137.36.34.bc.googleusercontent.com.https > kali.39342: UDP, l
```

11. Tráfico de una red completa

tcpdump net 192.168.43.0/24

```
(kali㉿kali)-[~]
└─$ sudo tcpdump net 192.168.0.0/24

tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
06:05:04.246315 IP 192.168.0.43.60684 > 239.255.255.250.1900: UDP, length 388
06:05:04.290739 IP 192.168.0.108.54350 > 239.255.255.250.1900: UDP, length 170
06:05:04.294040 IP 192.168.0.29.54354 > 239.255.255.250.1900: UDP, length 170
06:05:04.315826 IP kali.46647 > 254.red-80-58-61.staticip.rima-tde.net.domain: 4402
5.255.239.in-addr.arpa. (46)
06:05:04.338238 IP 254.red-80-58-61.staticip.rima-tde.net.domain > kali.46647: 4402
/0 (103)
06:05:04.339024 IP kali.56501 > 254.red-80-58-61.staticip.rima-tde.net.domain: 4328
60.100.in-addr.arpa. (42)
```

- ◆ Visualización de datos

12. Mostrar el contenido en ASCII

tcpdump lep -i eth0 -A

```
(kali㉿kali)-[~]
$ sudo tcpdump -i eth0 -A

tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
06:05:58.169222 ARP, Request who-has 192.168.1.163 tell 192.168.0.112, length 46
.....X.....f...p.....
06:05:58.212161 IP 192.168.0.34.51010 > 192.168.1.255.2000: UDP, length 4
E.. U.....a{ ... " .....B.....
06:05:58.252588 IP kali.35745 > 254.red-80-58-61.staticip.rima-tde.net.domain: 9518+ P
68.192.in-addr.arpa. (44)
E HX @ @ nP:= 5 40 % 163 1 168 192 in-addr arpa
```

13. Mostrar el contenido en hexadecimal

tcpdump -i eth0 -XX

```
(kali㉿kali)-[~]
$ sudo tcpdump -i eth0 -XX

tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
06:06:44.956616 IP 192.168.0.125.62524 > 192.168.1.255.2000: UDP, length 4
    0x0000: ffff ffff ffff 2c41 38ac 7d1c 0800 4500 .....A8.}...E.
    0x0010: 0020 fda8 0000 8011 b957 c0a8 007d c0a8 .....W...}..
    0x0020: 01ff f43c 07d0 000c d551 aaaa 0000 0000 ...<....Q.....
    0x0030: 0000 0000 0000 0000 0000 0000 .....
06:06:45.019552 IP kali.38188 > 254.red-80-58-61.staticip.rima-tde.net.domain: 16977+ PTR? 255.1.
168.192.in-addr.arpa. (44)
    0x0000: 00a0 26d2 689a 0800 27d1 f85d 0800 4500 ..&.h... '..]..E.
    0x0010: 0048 6b98 4000 4011 7fbe c0a8 006e 503a .Hk.@.@.....nP:
    0x0020: 3dfe 952c 0035 0034 4f94 4251 0100 0001 =..,.5.40.BQ....
    0x0030: 0000 0000 0000 0000 0332 3535 0131 0331 3638 .....255.1.168
    0x0040: 0331 3932 0769 6e2d 6164 6472 0461 7270 .192.in-addr.arp
    0x0050: 6100 000c 0001 .....
```

◆ Ejercicios combinados (operadores lógicos)

14. Tráfico desde una IP y en puerto 80

tcpdump -i eth0 src 192.168.43.143 and tcp port 80

```
(kali㉿kali)-[~]
$ sudo tcpdump -i eth0 src 192.168.0.110 and tcp port 80

tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
06:07:22.169933 IP kali.46762 > 82.221.107.34.bc.googleusercontent.com.http: Flags [S], seq 40127
59971, win 64240, options [mss 1460,sackOK,TS val 3172538922 ecr 0,nop,wscale 7], length 0
06:07:22.184712 IP kali.46762 > 82.221.107.34.bc.googleusercontent.com.http: Flags [.], ack 40190
42745, win 502, options [nop,nop,TS val 3172538937 ecr 3527942365], length 0
06:07:22.185282 IP kali.46762 > 82.221.107.34.bc.googleusercontent.com.http: Flags [P.], seq 0:31
0, ack 1, win 502, options [nop,nop,TS val 3172538937 ecr 3527942365], length 310: HTTP: GET /suc
cess.txt?ip=4 HTTP/1.1
06:07:22.202708 IP kali.46762 > 82.221.107.34.bc.googleusercontent.com.http: Flags [.] ack 217
```

15. Tráfico desde una IP excluyendo UDP

`tcpdump -i eth0 src 192.168.1.143 and not udp`

```
(kali㉿kali)-[~]
$ sudo tcpdump -i eth0 src 192.168.0.110 and not udp

tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
06:08:05.206588 IP kali.51522 > 191.144.160.34.bc.googleusercontent.com.https: Flags [S]
711525, win 64240, options [mss 1460,sackOK,TS val 696688775 ecr 0,nop,wscale 7], length 0
06:08:05.222314 IP kali.51522 > 191.144.160.34.bc.googleusercontent.com.https: Flags [.]
2160196, win 502, options [nop,nop,TS val 696688790 ecr 1968154662], length 0
```

16. Tráfico desde una IP en HTTP o HTTPS

`tcpdump -i eth0 src 192.168.1.143 and (port http or https)`

```
(kali㉿kali)-[~]
$ sudo tcpdump -i eth0 src 192.168.0.110 and port http or https

tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
06:09:10.319037 IP kali.56148 > 203.137.36.34.bc.googleusercontent.com.https: Flags [S], se
172758, win 64240, options [mss 1460,sackOK,TS val 2278237088 ecr 0,nop,wscale 7], length 0
06:09:10.334256 IP 203.137.36.34.bc.googleusercontent.com.https > kali.56148: Flags [S.], s
8257847, ack 3200172759, win 65535, options [mss 1412,sackOK,TS val 2541727260 ecr 22782370
,wscale 8], length 0
```

👉 Con esta lista de **16 ejercicios** tienes un recorrido completo:

- Empezando por capturar y guardar tráfico.
- Luego filtrando por protocolo, puerto, IP, MAC y red.
- Finalmente usando operadores lógicos y visualización en distintos formatos.